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Leveraging Deep Learning-based Approach for IoT Service Composition through Local Service Selection

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Abstract. The advent of the Internet of Things (IoT) has created opportunities for developing innovative smart systems, but it has also introduced complex requirements often subject to Quality of Service (QoS) constraints. Efficient service composition, essential for addressing these needs, involves selecting and integrating suitable services. Given the rapid expansion of IoT and the exponential increase in available services, the selection of IoT services poses significant challenges. In this context, several approaches have been proposed in the literature broadly categorized into two main types: global selection and local selection. Global selection approaches suffer from scalability issues, while local selection methods rely on local QoS constraints derived from a predefined set of candidate services, which is a limiting assumption in large and dynamic environments. In this paper, we present a novel deep learning-based QoS decomposition approach for IoT service local selection. Our method efficiently transforms global QoS constraints of complex needs, formalized as abstract processes, into distinct local constraints. This decomposition facilitates local selection without relying on a predefined set of services, enabling a flexible and dynamic selection process that accommodates the inherent variability and expansiveness of IoT environments. We have implemented and evaluated the proposed approach. The results show significant advancements in addressing the IoT service selection problem compared to the existing methods in the literature.

Keywords: Service composition, Local selection, QoS, Deep Learning, Attention Model, IoT

1 Introduction

In recent years, the concept of service has evolved from predominantly information systems to a dynamic landscape encompassing Internet of Things (IoT)-based systems. IoT is now present in various areas, spanning industries such as health and agriculture [7]. By seamlessly connecting devices, IoT enables the emergence of intelligent systems capable of gathering, processing, and analyzing real-time data, thereby facilitating data-driven decision-making. IoT plays a central role, particularly in fostering interconnection and collaboration among vari-

ous devices to effectively address the complex functional requirements of IoT applications. These requirements are typically specified as abstract processes, often integrating Quality of Service (QoS) constraints [5]. With the ever-growing number of available IoT services evolving within dynamic infrastructures, selecting an optimal QoS-aware service composition under end-to-end QoS requirements remains an open challenge.

In the literature, QoS-driven service selection is typically addressed through two main approaches: *global* and *local*. The first approach is based on a global search for the optimal service composition from an exhaustive set of all possible compositions [13, 12, 15]. Although effective in handling global QoS constraints, this approach suffers from scalability issues and becomes impractical in large and dynamic environments. Unlike the global approach, which considers dependencies among all selected services, the local approach offers the flexibility of independently selecting a service for each abstract task based on its local utility value [19, 6, 14, 20]. This method primarily relies on decomposing the global QoS constraints of a given abstract process into a series of local constraints associated with each abstract task. It enables local service selection while ensuring that the overall composition satisfies the global QoS constraints. By focusing on individual tasks and their local constraints, this approach achieves significant performance improvements while maintaining a high level of optimality. However, existing QoS decomposition approaches rely on a set of pre-discovered services [17, 14, 6]. The QoS values of this static set of services serve as the basis for defining the quality levels used to drive local constraints. As a result, the overall quality of the decomposition is intrinsically tied to the quality of these fixed services. If QoS-efficient services are overlooked in the pre-discovered set, it can significantly impair the decomposition process, potentially leading to suboptimal results or even failure.

To overcome these limitations, this paper introduces a novel attention-based deep learning global QoS decomposition approach for service composition. The primary objective is to derive local constraints from global QoS constraints within large-scale and dynamic systems. Unlike existing approaches, our method does not depend on the QoS values of a predefined set of services. Instead, it infers local QoS constraints by effectively capturing relevant task dependencies, historical patterns, and trends. These derived local QoS constraints enhance service selection, ensuring that relevant services are included and not overlooked during the selection process. This approach promotes more effective and efficient service composition compared to traditional methods.

To summarize, the contributions of the proposed approach can be outlined as follows:

- We introduce a predictive approach that leverages attention mechanisms and deep learning models to address the global QoS decomposition problem.
- Local service selection is performed based on inferred local QoS constraints, making our approach particularly suited for large-scale and dynamic systems where local constraints are derived independently of any predefined set of services.

- We have implemented and evaluated the proposed approach using a publicly available dataset, showcasing the effectiveness and efficiency of the proposed solution.

The remainder of this paper is organized as follows. Section 2 discusses related work. The proposed approach is presented in Section 3. Experimental results are discussed in Section 4. Finally, Section 5 concludes the paper.

2 Related Work

The service composition problem, initially explored in the context of Service-Oriented Architecture (SOA) and later extended to the Internet of Things (IoT) [12, 2], involves both functional attributes and QoS parameters to ensure that all tasks are completed while satisfying QoS constraints. This NP-Hard problem is often treated as a Multi-Choice Knapsack Problem [10]. Early approaches focused on linear optimization, such as linear integer programming [23], which effectively identifies optimal service compositions but is limited by computational intensity, making it impractical for large-scale service compositions.

To address the limitations of exact approaches, which suffer from scalability issues, global near-optimal approaches have been proposed [14, 6, 1]. In [13], the authors presented a Multi-objective Optimization-based meta-heuristic using NSGA II, which constructs the best solution from a population of initial solutions. Similarly, [12] proposed a genetic algorithm and Particle Swarm Optimization (PSO) approach that enhanced the diversity of the considered population. Further, [18] introduced an improved PSO version based on historical service composition solutions, while [11] integrated data mining clustering techniques with a genetic algorithm (GA) for global QoS-aware service composition in cloud computing. Other notable works include a hybrid Honey Bees Mating optimization algorithm for semantic Web Service Composition [9], and the Ant Lion optimization algorithm adapted to the composition problem [3]. Despite their performance improvements over exact methods, these approaches remain limited in tackling large-scale systems.

To overcome the limitations of global approaches, local approaches have been investigated. These approaches decompose global QoS constraints into a set of local constraints associated with the tasks of a given process. For each task, the appropriate service that fulfills the local constraints is selected. [2] proposed a Mixed Integer Programming (MIP)-based method for decomposing global constraints, transforming the value range of each QoS parameter into discrete quality levels used as candidates for local constraints. Similarly, [16] explored a Culture Genetic Algorithm (CGA)-based approach to define the local constraints, while [22] integrated user preferences into the decomposition process using Fuzzy logic technology and CGA. Additionally, [14] formulated the decomposition problem as a multi-objective optimization problem, solved using a multi-objective evolutionary algorithm (MOEA/D-IEpsilon). These approaches involve initially pre-discovering a static set of services based on criteria like offered functionalities, subsequently deriving local QoS constraints from various QoS quality

levels. However, this preliminary step disregards QoS values, and relevant services can be dismissed, limiting the applicability of these approaches, especially in dynamic IoT systems.

Other works have proposed reusing fragments of stored historical solutions to generate reusable service compositions [21]. However, such approaches often do not account for dependencies between tasks, reducing the quality of the generated solutions. More recent studies have used machine learning-based algorithms, such as Q-learning and Deep-Q-learning [8]. Although these approaches have made considerable improvements, they are limited within complex systems where the state space is large, making it challenging to effectively explore and identify high-quality solutions.

Given the aforementioned shortcomings, this paper investigates a novel deep-learning-enabled local selection approach. Specifically, we introduce an attention-based deep learning method for global QoS decomposition that captures task dependencies and historical patterns. This approach leverages local selection and attention mechanisms, coupled with deep learning models, presenting a significant advancement in addressing the complexities of service composition in dynamic large-scale IoT systems.

3 Attention-based deep learning-based Approach

Fig.1 shows the main components of the proposed approach. Given an abstract process specification with global QoS constraints, the first step consists of decomposing the global QoS constraints into a set of local constraints associated with each abstract task. Using the generated local constraints, a QoS-based local selection of services is performed. .

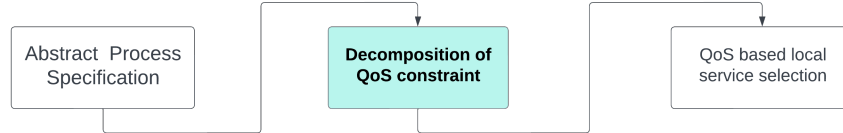


Fig. 1. Global overview of the proposed QoS decomposition-based approach

3.1 Problem formulation

An abstract process consists of multiple interconnected tasks, either composite or atomic, linked via operators like sequential (SEQ), parallel (AND), loop (Loop), or conditional (XOR). These structures can be converted into a sequential model,

as detailed in [4], so this work focuses on sequential cases. Global QoS constraints are associated with the process.

Definition 1. An abstract process is represented as a tuple $P = \{N, L, \gamma, \mathcal{Q}, W\}$ where:

- N : a set of nodes, either:
 - tasks $T = \{t_1, \dots, t_n\}$, where $t_i = \{nm_i, f_i\}$, with nm_i representing the task name and f_i the functionality.
 - process flow operators $O = \{O_1, \dots, O_g\}$, such as SEQ, AND, Loop, XOR.
- $L \subseteq N \times N$: set of edges.
- γ : a function mapping nodes to tasks or process flows.
- $\mathcal{Q} = \{Q_1, Q_2, \dots, Q_r\}$: global QoS constraints, where Q_j represents the j^{th} QoS parameter for the overall process.
- $W = \{w_1, w_2, \dots, w_r\}$: set of weight preferences for each QoS parameter, with $\sum_{j=1}^r w_j = 1$.

For an abstract process P with tasks $T = \{t_1, \dots, t_n\}$ and global QoS constraints $\mathcal{Q} = \{Q_1, Q_2, \dots, Q_r\}$, the decomposition problem involves predicting the percentage p_j^i that each task t_i consumes in each QoS constraint Q_j , with $\sum_{j=1}^r p_j^i = 1$ for every QoS constraint Q_j .

The amount of QoS Q_j allocated to each task is $e_j^i = p(e_j^i) \times Q_j$, ensuring local selection aligns with global QoS constraints.

3.2 Global QoS decomposition

Our decomposition approach combines the power of deep learning models and attention mechanisms. The idea entails predicting a percentage contributed by each task to each global QoS parameter. As depicted in Fig.2, the approach we propose involves three parts:

Input Embedding. The specification of a process P is represented as a vector of task identifiers. The global QoS constraints and the vector of task identifiers are used to derive local constraints from the predicted percentage.

The first step in the model is generating an encoded vector for each task in the process, produced by the Embedding Layer. This layer ensures an accurate representation of each task, learned during training, and enables capturing task similarities, which methods like OneHotEncoding fail to achieve.

Attention Encoder The aim of this model component, utilizing an Attention Encoder, is to capture long-range dependencies and complex patterns from the task representation input. This input is specified as a vector sequence that represents all tasks.

The attention mechanism, based on the Query-Key-Value (QKV) model, processes the general context of the entire sequence within each vector. The attention encoder outputs a dynamically enriched task vector sequence of the same

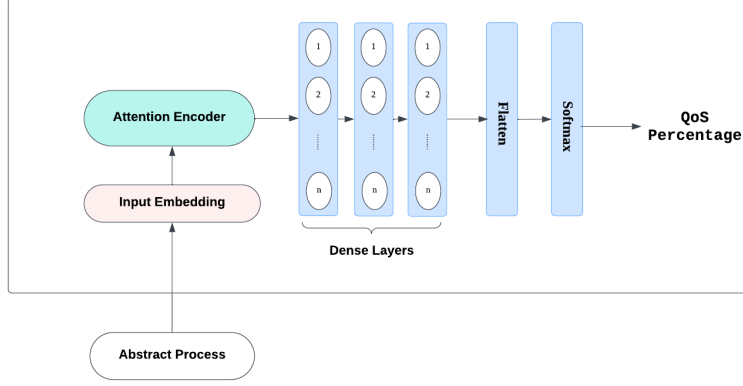


Fig. 2. Architecture of the proposed attention-based deep learning decomposition approach

length, capturing relevant task information and dependencies. This is formulated as:

$$Attention(Q, K, V) = softmax \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

The input sequence T is transformed into query (Q), key (K), and value (V) matrices with dimensions d_k and d_v :

$$Q = W_q \times T, \quad K = W_k \times T, \quad V = W_v \times T$$

where W_q , W_k , and W_v are the learned weight matrices.

QoS probability prediction. In this final step, each new representation generated by the attention encoder is passed through dense layers. The primary purpose is to extract comprehensive information from the final representation vector to produce a singular and aggregated value for each task. These values are then grouped into a vector and forwarded to the *softmax function*, given in Equation 2, to transform the output of the dense layers into a list of task-specific QoS percentages required for each task.

$$\sigma(t_i) = \frac{e^{t_i}}{\sum_{j=1}^n e^{t_j}} \quad for \ i = 1, \dots, n \quad (2)$$

The values of the output vector are p_j^i with $1 \leq i \leq n, 1 \leq j \leq r$ which corresponds to the proportion of the j^{th} QoS for the i^{th} task. The local QoS values are then calculated by simply multiplying each generated percentage by the global QoS value.

$$e_j^i = p_j^i \times Q_j \quad (3)$$

where e_j^i is the local constraint of the j^{th} global constraint Q_j for the i^{th} task t_i . Here, p_j^i is the predicted percentage of the j^{th} global constraint Q_j for the i^{th} task.

Definition 2. *Local QoS constraints*

Given an abstract process $P = \{N, L, \gamma, Q, W\}$, the result of the global QoS decomposition is a set of local constraints associated to tasks $T = \{t_1, \dots, t_n\}$, where $t_i = \{nm_i, f_i, \mathcal{E}_i\}$ with $\mathcal{E}_i = \{e_1^i, \dots, e_r^i\}$, such as $e_j^i (1 \leq j \leq r, 1 \leq i \leq n)$ is the local constraint of the j^{th} global constraint Q_j of the i^{th} task such as $Agg(e_j^i) \leq Q_j$, where $Agg(e_j^i)$ is the aggregation function.

3.3 Local service selection

The services search space can be refined and pruned using inferred local QoS constraint values. This refinement can leverage existing functional-based discovery approaches [14, 1] by integrating derived local constraints specific to each task. Services that do not meet these local constraints can then be systematically filtered out from consideration.

$$\begin{cases} q_{jk}^i \leq e_j^i & \text{For negative attributes} \\ q_{jk}^i \geq e_j^i & \text{For positive attributes} \end{cases} \quad (4)$$

where e_j^i is the inferred local constraint of the j^{th} global constraint of the i^{th} task and q_{jk}^i is the quality value offered by the k^{th} service s_{i_k} for the i^{th} task for the j^{th} constraint with $1 \leq i \leq n, 1 \leq j \leq r, 1 \leq k \leq h$.

The goal is to select services that maximize utility while satisfying global QoS constraints. The Simple Additive Weighting (SAW) technique is used to map a service's multi-dimensional QoS vector into a single utility value. The distance between a QoS value q_{jk}^i and the predicted QoS constraint value e_j^i

$$q_{dist_{jk}}^i = \begin{cases} \max(\frac{e_j^i - q_{jk}^i}{e_j^i}, 0) & \text{For negative attributes} \\ \max(\frac{q_{jk}^i - e_j^i}{q_{jk}^i}, 0) & \text{For positive attributes} \end{cases} \quad (5)$$

The utility of a service s_{i_k} for a task t_i is calculated by summing the weighted QoS distances across all attributes: 6.

$$U(s_{i_k}) = \sum_{j=1}^r w_j \cdot q_{dist_{jk}}^i \quad (6)$$

where w_j with $(1 \leq j \leq r)$ is the weight preference for the j^{th} QoS parameter, $w_i \in [0, 1]$, and $\sum_{j=1}^r w_j = 1$.

Finally, for a task t_i , a service s_{i_k} that respects local constraints and maximizes utility is selected.

$$\begin{cases} \max(U(s_{i_k})) \\ q_{jk}^i \leq e_j^i & \text{For negative attributes} \\ q_{jk}^i \geq e_j^i & \text{For positive attributes} \end{cases} \quad (7)$$

4 Experimental evaluation

4.1 Experimental setup

Used dataset

Experiments were conducted using the WSDream dataset, which includes 1,974,675 records of service invocations from 5,825 services and 339 users. The dataset was transformed to define tasks, each associated with services providing different QoS values.

Optimization and Loss Function

To enhance the model's performance and assess its effectiveness, selecting an appropriate loss function and optimization method is essential. Since QoS values form a probability distribution, Cross Entropy Loss was chosen to compare predicted and actual distributions. Additionally, the Adam optimization method was employed for its ability to achieve faster convergence with minimal hyperparameter tuning, improving training efficiency.

$$Loss = - \sum_{j=1}^N p_j^i \log(\hat{p}_j^i) \quad (8)$$

where $\hat{p}_j^i$ and p_j^i are the predicted and ground truth value of the percentage of j^{th} QoS parameter of the i^{th} task, N is the number of tasks.

4.2 Comparative study

We have conducted a comprehensive comparative study to evaluate the effectiveness of our approach compared to the following existing approaches:

- *MOEA (Multi-Objective Evolutionary-based Approach)* [14]: Uses a Multi-Objective Evolutionary Genetic Algorithm for near-optimal decomposition of global QoS constraints.
- *AQCD(Adaptive Global QoS Constraints Decomposition)* [22]: Employs Fuzzy logic and Cultural Genetic Algorithms for near-optimal local constraints.

Fig.3 and Fig.4 show, respectively, a comparison of the execution time and the optimality of our decomposition-based approach with the approaches MOEA and AQCD. The optimality is computed by comparing the overall utility value of the solution found by each approach (U_{sol}) with the utility value of the optimal

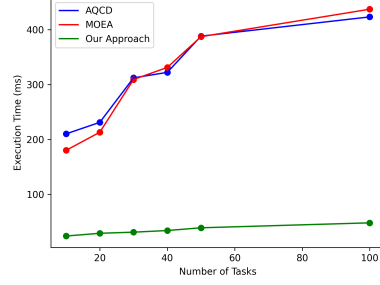


Fig. 3. Evaluation of the execution time according to the number of tasks

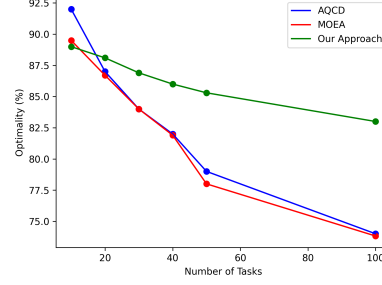


Fig. 4. Variation of the optimality according to the number of tasks

solution obtained from the absolute approach (which consists of exploring all possible combinations and finding the best combination) U_{abs} : $optimality = U_{sol}/U_{abs}$.

Our approach stands out with interesting performance compared to the existing approaches, especially in terms of execution time, while maintaining a high level of optimality. Although the AQCD and MOEA approaches achieve better optimality for small-scale problems, they encounter significant runtime degradation and a rapid decrease in optimality as the number of tasks increases. In contrast, our approach maintains a stable execution time, while experiencing a gradual decrease in optimality. This offers a well-balanced trade-off between performance and scalability.

The second part of the experiments involves evaluating the performance of the different approaches according to the number of available services.

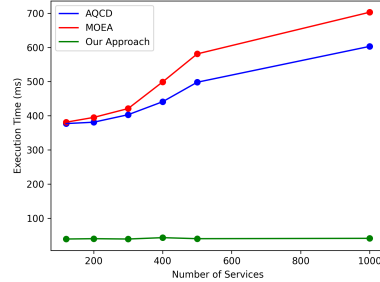


Fig. 5. Evaluation of the execution time according to the number of services

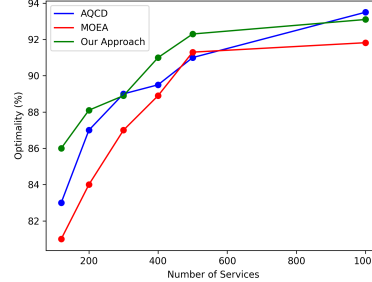


Fig. 6. Variation of the optimality according to the number of services

In Fig.6, we can see that the execution time of both MOEA and AQCD approaches increases significantly as the number of services expands. This escalation is attributed to the local decomposition in these approaches, which relies on the different quality of service parameters derived from service QoS values. Consequently, as the number of services grows, the local decomposition time also escalates accordingly. In contrast, our approach remains unaffected by the number of services. Decomposition is based solely on tasks and their relationships, significantly enhancing the efficiency of the composition approach.

5 Conclusion

In this paper, we introduced a novel solution, a predictive global QoS decomposition approach designed to address the local QoS-based service selection problem. Our approach harnesses the power of deep learning-based models and attention mechanisms, which have demonstrated effectiveness across various domains in enhancing the deep learning model's performance. Unlike existing works, our approach does not rely on a static set of predefined services to infer the local constraints. Instead, it dynamically captures task relationships from task specifications and past experiences, ensuring an adaptive decomposition process. Through comprehensive experiments, we have demonstrated the effectiveness of the proposed approach sticking a balance between performance and optimality. This makes it promising for addressing the composition challenges in large-scale real-time IoT systems.

We plan to extend our approach to handle timed-constrained processes. Beyond structural task dependencies, abstract processes often involve timed constraints among tasks, significantly increasing the complexity composition. This extension will enable to tackle a broader range of IoT applications, particularly those requiring time-sensitive tasks. Additionally, we aim to incorporate online dynamic adjustment of local constraints to adapt to changes in the environment.

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